

## PHUKET, Thailand

WMO: 485640

Lat: 7.883N

Lon: 98.400E

Elev: 13

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |       |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|-------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |       |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 10            | 75.0       | 75.7 | 65.8                          | 95.1 | 84.2 | 67.3 | 100.2 | 84.0 | 11.1                  | 85.3 | 9.9 | 84.9 | 1.5                   | 300  | 0.357 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 11.7                   | 94.7            | 79.1 | 93.5 | 78.9 | 92.5 | 78.8 | 81.8                | 90.2 | 81.1 | 89.2 | 80.7 | 88.6 | 4.7                  | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.8                            | 154.6 | 85.5 | 79.1 | 151.2 | 84.9 | 78.6 | 148.8 | 84.6 | 45.6          | 90.7 | 44.9 | 89.5 | 44.3 | 88.8 | 87.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |  |  |
| 10.0              | 9.3  | 8.1 | 72.9                       | 96.6 | 1.5                | 1.7 | 71.8                                               | 97.8 | 70.9       | 98.8 | 70.1       | 99.7 | 69.0       | 101.0 |  |  |
|                   |      |     | 68.4                       | 83.5 | 2.3                | 1.3 | 66.8                                               | 84.5 | 65.4       | 85.2 | 64.2       | 86.0 | 62.5       | 87.0  |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 84.0   | 83.5  | 84.9  | 85.8  | 85.9  | 85.0  | 84.2  | 83.9  | 83.6  | 82.8  | 82.6  | 83.1  | 83.0  |
|                                                                        | DBStd       | 2.28   | 1.69  | 1.61  | 2.02  | 2.08  | 2.27  | 2.13  | 2.12  | 2.35  | 2.12  | 1.99  | 1.76  | 1.71  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 12422  | 1039  | 978   | 1111  | 1077  | 1084  | 1026  | 1050  | 1043  | 985   | 1011  | 994   | 1024  |
|                                                                        | CDD65       | 6947   | 574   | 558   | 646   | 627   | 619   | 576   | 585   | 578   | 535   | 546   | 544   | 559   |
|                                                                        | CDH74       | 82117  | 6606  | 6862  | 8283  | 8143  | 7735  | 6941  | 6917  | 6710  | 5857  | 5823  | 6036  | 6204  |
| (13)                                                                   | CDH80       | 33418  | 2565  | 3042  | 3970  | 3929  | 3446  | 2865  | 2753  | 2590  | 2011  | 1939  | 2113  | 2195  |
| (14)                                                                   | Wind        | WSAvg  | 2.3   | 2.4   | 2.4   | 2.0   | 1.7   | 2.0   | 2.4   | 2.7   | 2.9   | 2.4   | 1.9   | 2.6   |
| (15) Precipitation                                                     | PrecAvg     | 92.5   | 1.7   | 0.9   | 2.0   | 4.9   | 13.7  | 12.0  | 10.9  | 10.6  | 16.1  | 12.3  | 6.9   | 2.5   |
|                                                                        | PrecMax     | 109.7  | 5.4   | 6.8   | 7.8   | 12.4  | 27.2  | 21.4  | 18.5  | 21.5  | 20.0  | 28.1  | 12.2  | 5.0   |
|                                                                        | PrecMin     | 71.7   | 0.0   | 0.0   | 0.0   | 0.1   | 5.4   | 3.7   | 4.5   | 3.9   | 8.2   | 5.4   | 3.7   | 0.2   |
|                                                                        | PrecStd     | 12.0   | 1.7   | 1.7   | 2.0   | 3.3   | 5.5   | 5.1   | 4.6   | 4.7   | 3.6   | 6.3   | 3.0   | 1.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 93.1  | 95.2  | 96.3  | 96.5  | 95.0  | 93.4  | 92.6  | 93.0  | 92.3  | 92.0  | 92.1  | 92.0  |
|                                                                        |             | MCWB   | 78.0  | 77.7  | 79.1  | 80.1  | 80.4  | 79.7  | 79.1  | 78.9  | 78.9  | 79.1  | 79.1  | 78.5  |
|                                                                        | 2%          | DB     | 91.5  | 93.4  | 94.7  | 94.5  | 93.1  | 91.8  | 91.1  | 91.3  | 90.5  | 90.4  | 90.7  | 90.5  |
|                                                                        |             | MCWB   | 77.2  | 77.3  | 78.6  | 79.9  | 80.1  | 79.5  | 78.8  | 78.8  | 78.8  | 78.9  | 78.8  | 78.0  |
|                                                                        | 5%          | DB     | 90.3  | 92.1  | 93.2  | 93.0  | 91.8  | 90.7  | 90.0  | 90.0  | 89.4  | 89.2  | 89.6  | 89.3  |
|                                                                        |             | MCWB   | 77.0  | 77.1  | 78.3  | 79.6  | 79.9  | 79.4  | 78.8  | 78.6  | 78.7  | 78.7  | 78.4  | 77.6  |
|                                                                        | 10%         | DB     | 89.0  | 90.8  | 91.8  | 91.5  | 90.4  | 89.5  | 88.9  | 88.8  | 88.1  | 87.9  | 88.3  | 88.2  |
|                                                                        |             | MCWB   | 76.7  | 76.9  | 78.1  | 79.4  | 79.7  | 79.4  | 78.7  | 78.5  | 78.5  | 78.5  | 78.2  | 77.3  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 80.2  | 80.7  | 81.9  | 82.8  | 82.5  | 81.8  | 81.0  | 81.0  | 80.9  | 81.0  | 80.8  | 80.9  |
|                                                                        |             | MCDB   | 89.6  | 90.5  | 92.3  | 92.2  | 91.0  | 89.0  | 88.5  | 88.2  | 88.1  | 88.1  | 88.7  | 88.7  |
|                                                                        | 2%          | WB     | 79.1  | 79.5  | 80.8  | 81.7  | 81.6  | 81.0  | 80.2  | 80.1  | 80.0  | 80.0  | 79.9  | 79.6  |
|                                                                        |             | MCDB   | 87.9  | 89.1  | 90.7  | 90.6  | 89.7  | 88.1  | 87.5  | 87.4  | 87.2  | 87.0  | 87.6  | 87.4  |
|                                                                        | 5%          | WB     | 78.3  | 78.8  | 80.0  | 80.9  | 80.9  | 80.5  | 79.8  | 79.6  | 79.4  | 79.4  | 79.2  | 78.8  |
|                                                                        |             | MCDB   | 86.7  | 88.2  | 89.4  | 89.4  | 88.6  | 87.5  | 86.9  | 86.8  | 86.3  | 86.2  | 86.5  | 86.5  |
|                                                                        | 10%         | WB     | 77.8  | 78.1  | 79.3  | 80.3  | 80.4  | 79.9  | 79.2  | 79.1  | 78.9  | 79.0  | 78.8  | 78.2  |
|                                                                        |             | MCDB   | 85.9  | 87.3  | 88.3  | 88.4  | 87.8  | 86.7  | 86.0  | 86.1  | 85.6  | 85.6  | 85.8  | 85.5  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.9  | 13.2  | 12.9  | 11.7  | 10.6  | 10.1  | 9.9   | 9.7   | 10.0  | 10.2  | 10.3  | 10.7  |
|                                                                        |             | MCDBR  | 13.3  | 14.6  | 14.3  | 13.0  | 12.2  | 11.6  | 11.1  | 11.3  | 11.9  | 12.2  | 12.1  | 12.2  |
|                                                                        |             | MCWBR  | 4.0   | 4.4   | 4.2   | 3.6   | 3.3   | 3.3   | 3.2   | 3.1   | 3.5   | 3.7   | 3.5   | 3.8   |
|                                                                        | 5% WB       | MCDBR  | 12.2  | 13.4  | 13.3  | 12.0  | 11.1  | 10.6  | 10.3  | 10.4  | 10.9  | 11.2  | 11.2  | 11.2  |
|                                                                        |             | MCWBR  | 4.1   | 4.6   | 4.4   | 4.0   | 3.7   | 3.8   | 3.7   | 3.6   | 3.8   | 3.9   | 3.7   | 4.0   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.460 | 0.479 | 0.494 | 0.477 | 0.454 | 0.457 | 0.459 | 0.451 | 0.437 | 0.442 | 0.449 | 0.456 |
|                                                                        | taud        |        | 2.328 | 2.246 | 2.245 | 2.301 | 2.406 | 2.410 | 2.388 | 2.404 | 2.439 | 2.425 | 2.389 | 2.351 |
|                                                                        | Ebn at Noon |        | 272   | 270   | 266   | 266   | 267   | 263   | 264   | 270   | 278   | 277   | 272   | 270   |
|                                                                        | Edh at Noon |        | 41    | 46    | 46    | 43    | 38    | 37    | 38    | 38    | 37    | 38    | 38    | 39    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1725  | 1986  | 1994  | 1882  | 1606  | 1557  | 1546  | 1576  | 1544  | 1524  | 1526  | 1530  |
|                                                                        | RadStd      |        | 113   | 104   | 180   | 123   | 119   | 115   | 93    | 91    | 77    | 132   | 105   | 134   |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only            | +0.63 | N/A     | N/A    | +1.48   | +0.95 | +0.94 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (12 neighbors) | +0.36 | +0.56   | N/A    | N/A     | +0.36 | +0.41 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page